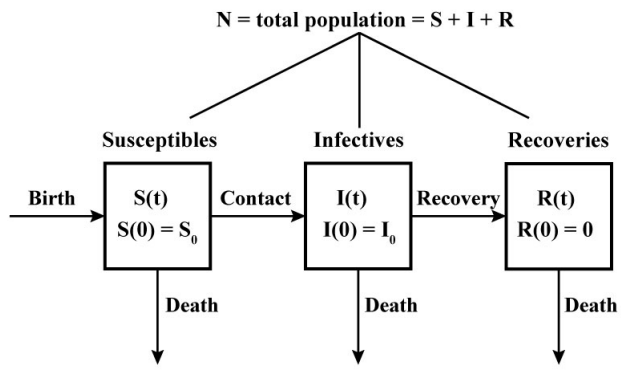
# Phase 4 (part 3): 3-D Systems and $2^{nd}$-Order Equations

1.] An SIR model is a system of three differential equations that models the spread of an infectious disease throughout a population. Consider the compartment based model below and the associated system of equations.

(a) The classic endemic SIR model can be described by the following diagram:




We can assemble the following equations based on this diagram:

$$\frac{dS}{dt} = \alpha N - \beta S\frac{I}{N} - \mu S$$
$$\frac{dI}{dt} = \beta S\frac{I}{N} - \gamma I - \mu I$$
$$\frac{dR}{dt} = \gamma I - \mu R$$

Here, $S_0$ and $I_0$ are the initial number of susceptible and infected persons at the beginning and the parameters $\alpha$, $\beta$, $\gamma$, and $\mu$ are the birth, incidence, recovery, and death rates, respectively. $N$ is the total number of people in the population assumed constant. Solve this system for the following parameter values:

$$N = 303824640$$
$$\beta = 0.32851429$$
$$\mu = 0.00872$$
$$\alpha = 0.00872$$
$$\gamma = 0.11673$$

$R_0 = \frac{\beta}{\gamma} = 2.814$

with initial conditions, $I_0 = 1$, $R_0 = 0$, and $S_0 = N - 1$, which means there is only one infected person in the whole population. Once you solve the equations, plot the solutions on the same figure and change the incidence rate, $\beta$, to see what happens.

(b) Solve the system again but make the incidence rate $\beta$ a function of time of the form

$$\beta(t) = b_0 + b_1 \sin(\pi t/6).$$

This simulates a periodic infectious rate that may change throughout a year. It is up to you to decide the parameter values for $b_0$ and $b_1$.

2.] The damped mass-on-a-spring system is modeled by the second-order differential equation

$$m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = kf(t),$$

where $m > 0$ is the mass, $b \geq 0$ is the damping coefficient, $k > 0$ is the spring constant, $f(t)$ is a forcing function, and $y(t)$ is the displacement from equilibrium. This equation describes oscillatory motion where the damping force opposes the velocity, causing the oscillations to gradually decrease in amplitude over time. Write this system as a first-order system of ODEs, then solve it in Python for various values of $m, b$, and $k$ and for different forcing functions $f(t)$.

Define $v = \frac{dy}{dt}$ be the velocity of the mass, then

$\frac{dv}{dt} = \frac{d^2y}{dt^2}$, which means

$$m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = kf(t)$$

$$\Rightarrow \quad m\frac{dv}{dt} + bv + ky = kf(t)$$

$$\Rightarrow \quad \frac{dv}{dt} = -\frac{k}{m}y - \frac{b}{m}v + \frac{k}{m}f(t)$$

This gives the following first order system:

$$\begin{cases} \frac{dy}{dt} = v & y(0) = y_0 \\ \frac{dv}{dt} = -\frac{k}{m}y - \frac{b}{m}v + \frac{k}{m}f(t) & v(0) = v_0 \end{cases}$$